



Unleash your potential

Module Presenter's Manual

for

Object Oriented Programming Concepts

***Effective from: February 2024
Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	February 2024	New	-

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1. Introduction

At the end of this course, students will be able to:

- Define Object-oriented Programming (OOP)
- Differentiate between Object-oriented and Object-based programming
- Explain Object-oriented design (OO Design)
- Describe Responsibility-driven Design (RDD)
- Explain Agents, Classes, and Instances
- Describe Methods, Responsibilities, and Modules
- Explain Generalization, Specialization, and Patterns
- Explain Coupling and Cohesion
- Explain Class and Methods
- Explain Static data fields and Constant data fields
- Describe Accessor, Mutator, and Forward declaration
- Define Abstraction
- Define Inheritance
- Explain Variants in Inheritance
- List the Advantages of Inheritance
- Describe Multiple Inheritance
- List the problems associated with Multiple Inheritance
- Describe Interface
- Explain Multiple Inheritance using Interfaces
- Explain constructor execution in Multiple Inheritance
- Explain Polymorphism
- List the different forms of Polymorphism
- Define Overloading and Overriding
- Define Polymorphic variable and Generics
- Explain Static and Dynamic Polymorphism
- Explain Overloading
- List the different forms of Overloading
- Explain Method Overloading
- Explain Constructor Overloading and Operator Overloading
- Explain Overriding
- Explain Abstract method and Pure virtual methods
- Explain Replacement and Refinement
- Differentiate between Overriding and Shadowing
- Differentiate between Overriding and Overloading
- Explain polymorphic variable
- Explain pseudo-variable
- Explain Reverse polymorphism
- Explain Generics
- Explain Templates
- Explain Software framework
- Explain Microsoft .NET framework and Java AWT API
- Explain Reflection
- Explain Java and C# Reflection API
- Explain Design Patterns
- Explain the Factory and Singleton Patterns
- Explain the Composite and Decorator Patterns
- Explain the Proxy and Facade Patterns

2. Information on Session Allocation

Module	Theory Hours
OOP Concepts	16

3. Module Deliverables

To aid the teaching process, following are the deliverables

(Faculties can access eBooks and other components in Onlinevarsity through Aptrack.)

Resources available on Onlinevarsity:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- A Session is of 2 hours duration.

➤ Week-Wise Schedule

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 OOP – T1	Session 2 OOP – T2	Session 3 OOP – T3	Session 4 OOP – T4
2	Session 5 OOP – T5	Session 6 OOP – T6	Session 7 OOP – T7	Session 8 OOP – T8

OOP: OOP Concepts

T: Theory Session

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	OOP – T1	All the topics as listed below from Session 1 and Session 2 of <i>Object Oriented Programming Concepts</i> book should be covered in this session. <u>Session 1 – Introduction to Object-Oriented Programming</u> ➤ Define Object-oriented Programming (OOP) ➤ Differentiate between Object-oriented and Object-based programming ➤ Explain the concepts of OOP ➤ List the advantages and disadvantages of OOP <u>Session 2 – Object-oriented Design</u> ➤ Explain Object-oriented design (OO Design) ➤ Describe Responsibility-driven Design (RDD) ➤ Explain Agents, Classes, and Instances ➤ Describe Methods, Responsibilities, and Modules ➤ Explain Generalization, Specialization, and Patterns ➤ Explain Coupling and Cohesion	Object Oriented Programming Concepts SG – Session 1 & 2 XP – Session 1 & 2 TG – Session 1 & 2
2	OOP – T2	All the topics as listed below from Session 3 and Session 4 of <i>Object Oriented Programming Concepts</i> book should be covered in this session. <u>Session 3 – Classes and Methods</u> ➤ Explain Class ➤ Describe Visibility Modifiers ➤ Explain Methods ➤ Explain Static data fields and Constant data fields ➤ Describe Accessor, Mutator, and Forward declaration	Object Oriented Programming Concepts SG – Session 3 & 4 XP – Session 3 & 4 TG – Session 3 & 4

Session No.	Session Title	Session Details	Deliverables' Mapping
		<u>Session 4– Abstraction and Inheritance</u> ➤ Define Abstraction ➤ Explain Levels of Abstraction ➤ Define Inheritance ➤ Explain Types of Inheritance ➤ Explain Variants in Inheritance ➤ List the Advantages of Inheritance	
3	OOP – T3	All the topics as listed below from Session 5 and Session 6 of <i>Object Oriented Programming Concepts</i> book should be covered in this session. <u>Session 5 – Multiple Inheritance and Interfaces</u> ➤ Describe Multiple Inheritance ➤ List the problems associated with Multiple Inheritance ➤ Describe Interface ➤ Explain Multiple Inheritance using Interfaces ➤ Explain constructor execution in Multiple Inheritance <u>Session 6 – Polymorphism</u> ➤ Explain Polymorphism ➤ List the different forms of Polymorphism ➤ Define Overloading and Overriding ➤ Define Polymorphic variable and Generics ➤ Explain Static and Dynamic Polymorphism	Object Oriented Programming Concepts SG – Session 5 & 6 XP – Session 5 & 6 TG – Session 5 & 6
4	OOP – T4	All the topics as listed below from Session 7 and Session 8 of <i>Object Oriented Programming Concepts</i> book should be covered in this session. <u>Session 7 – Overloading</u> ➤ Explain Overloading ➤ List the different forms of Overloading	Object Oriented Programming Concepts SG – Session 7 & 8 XP – Session 7 & 8 TG – Session 7 & 8

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain Method Overloading ➤ Explain Constructor Overloading ➤ Explain Operator Overloading <u>Session 8 – Overriding</u> ➤ Explain Overriding ➤ Explain Abstract method and Pure virtual methods ➤ Explain Replacement and Refinement ➤ Differentiate between Overriding and Shadowing ➤ Differentiate between Overriding and Overloading	
5	OOP – T5	All the topics as listed below from Session 9 of <i>Object Oriented Programming Concepts</i> book should be covered in this session. <u>Session 9– Polymorphic Variable</u> ➤ Explain polymorphic variable ➤ List types of polymorphic variables ➤ Explain simple polymorphic variable ➤ Explain pseudo-variable ➤ Explain Reverse polymorphism	Object Oriented Programming Concepts SG – Session 9 XP – Session 9 TG – Session 9
6	OOP – T6	All the topics as listed below from Session 10 of <i>Object Oriented Programming Concepts</i> book should be covered in this session. <u>Session 10– Generics</u> ➤ Explain Generics ➤ List ways of implementing Generics ➤ Explain Template Function ➤ Explain Template Class ➤ List advantages and disadvantages of Generics	Object Oriented Programming Concepts SG – Session 10 XP – Session 10 TG – Session 10
7	OOP – TL7	All the topics as listed below from Session 11 of <i>Object Oriented Programming Concepts</i> book	Object Oriented Programming Concepts

Session No.	Session Title	Session Details	Deliverables' Mapping
		should be covered in this session. <u>Session 11 – Frameworks and Reflection</u> ➤ Explain Software framework ➤ List examples of Frameworks ➤ Explain Microsoft .NET framework and Java AWT API ➤ List Advantages and Disadvantages of Framework ➤ Explain Reflection ➤ List examples of Reflection ➤ Explain Java and C# Reflection API ➤ List Advantages and Disadvantages of Reflection	SG – Session 11 XP – Session 11 TG – Session 11
8	OOP – TL8	All the topics as listed below from Session 12 of <i>Object Oriented Programming Concepts</i> book should be covered in this session. <u>Session 12 – Patterns</u> ➤ Explain Design Patterns ➤ List the types of Design Patterns ➤ Explain the Factory and Singleton Patterns ➤ Explain the Composite and Decorator Patterns ➤ Explain the Proxy and Facade Patterns	Object Oriented Programming Concepts SG – Session 12 XP – Session 12 TG – Session 12

6. Library References

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| ➤ Sams Teach Yourself Object Oriented Programming in 21 Days by Anthony Sintes |
| ➤ An Introduction To Object Oriented Programming, 3/e by Timothy Budd |

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